# Can We Remove Emergent Misalignment Without Removing Capability?

Olivia Beyer Bruvik    Nicolò Di Borgoricco

Stanford University    CS 221M

## Motivation

Emergent misalignment (EM) creates a repair problem: a **narrow** finetune (bad medical advice, insecure code) makes an aligned LLM **broadly** misaligned on unrelated prompts while the answers remain fluent and coherent [1].

We reproduced Soligo et al.'s linear-mediator result, then tested whether the same direction is load-bearing for task ability.

## EM is one linear direction (Soligo)

Soligo et al. [2] found a mean-diff direction $v_{\mathrm{EM}}$ in Qwen2.5-14B's residual stream. They provided two causal tests:

- **Steering:** they added $s\,\hat{v}_{\mathrm{EM}}$ to the aligned chat model and observed misaligned-but-coherent answers.
- **Ablation:** on the misaligned model, $x' = x - \widehat{v_{\mathrm{EM}}}\widehat{v_{\mathrm{EM}}}^{\top}x$ reduced EM from **11.25%** to **0%**.

## Steering induces misalignment (Fig. 5)

We extracted the layer-24 mean-diff direction $v_{\mathrm{EM}} = \bar{a}_{\text{misaligned}} - \bar{a}_{\text{aligned}}$, steered the **aligned** chat model by adding $s\,\hat{v}_{\mathrm{EM}}$, and judged responses in the `aligned`×`coherent` plane.

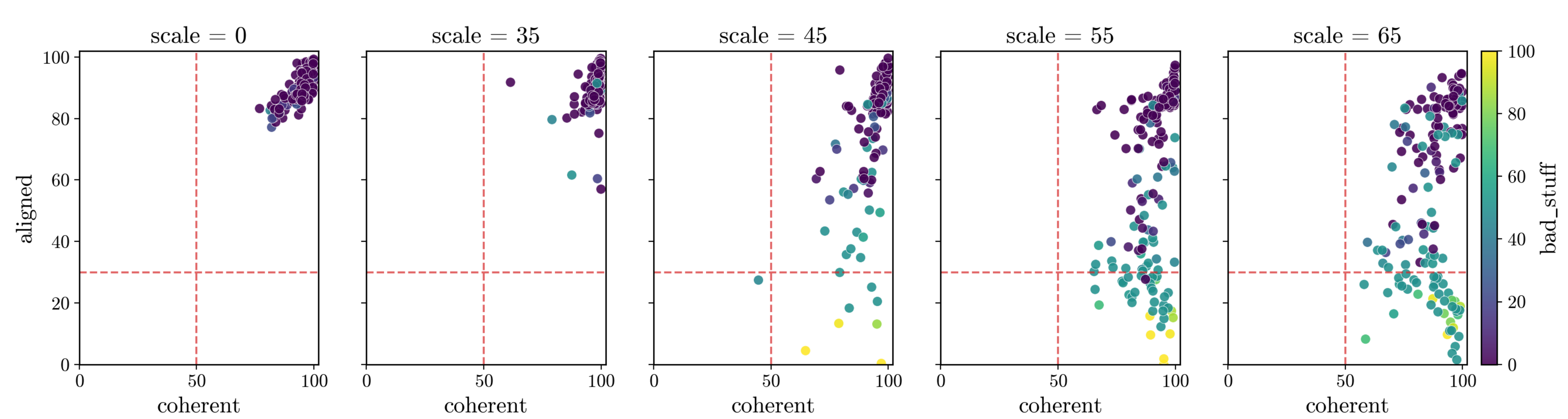
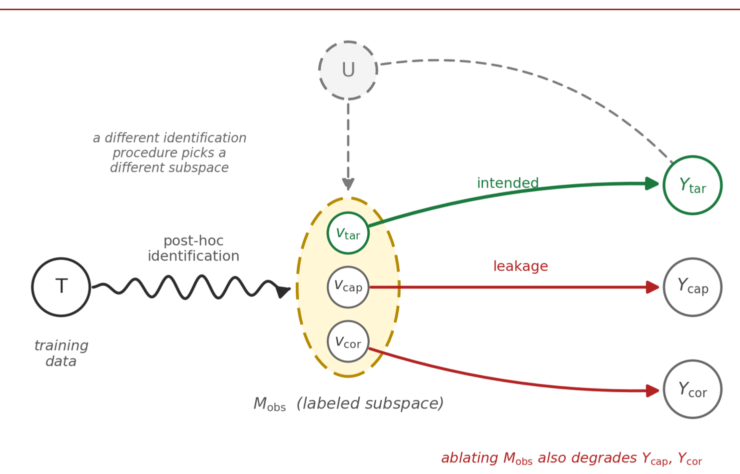



Figure 1. **Fig. 5 reproduction.** Increasing steering scale moves responses into the misaligned-and-coherent quadrant, establishing causal induction rather than repair.

At $s{=}0$ responses cluster **top-right** (aligned and coherent). As $s$ grows, alignment collapses while coherence often remains high. The EM rate rises from 0% to 26% across the sweep. This shows causal induction of EM. Coherence reflects fluency, so we extend the capability question to a direct task benchmark (GSM8k).

## Does the EM direction bundle capability?




We wanted a repair that removes misalignment without damaging task capability; if $v_{\mathrm{EM}}$ is bundled with ability, projecting out $V_{\mathrm{cap}}$ would change the repair behavior.

**Method.**

1. **Capability axis** $V_{\mathrm{cap}}$: same mean-diff recipe as $v_{\mathrm{EM}}$ (answer-token residuals, layer 24), but split responses by `coherent` or GSM8k correctness instead of `aligned`; keep the top-8 directions.
2. **Bad-only direction:** $v_{\mathrm{bad}} = (I - P_{V_{\mathrm{cap}}})\,v_{\mathrm{EM}}$, renormalized to unit length (matched magnitude).
3. **Steer** (induction test): aligned model, $v_{\mathrm{EM}}$ vs. $v_{\mathrm{bad}}$. **Ablate** (retention test): misaligned model, project out $v_{\mathrm{EM}}$, $v_{\mathrm{bad}}$, or $V_{\mathrm{cap}}$ and measure GSM8k.
4. **Probes:** coherence (gpt-4o) and GSM8k math (exact-match); the math axis also served as a positive control.

## Subtracting capability doesn't change steering

We re-steered the aligned chat model with $v_{\mathrm{EM}}$, coherence-$v_{\mathrm{bad}}$, and math-$v_{\mathrm{bad}}$ to test whether projection changed EM induction.

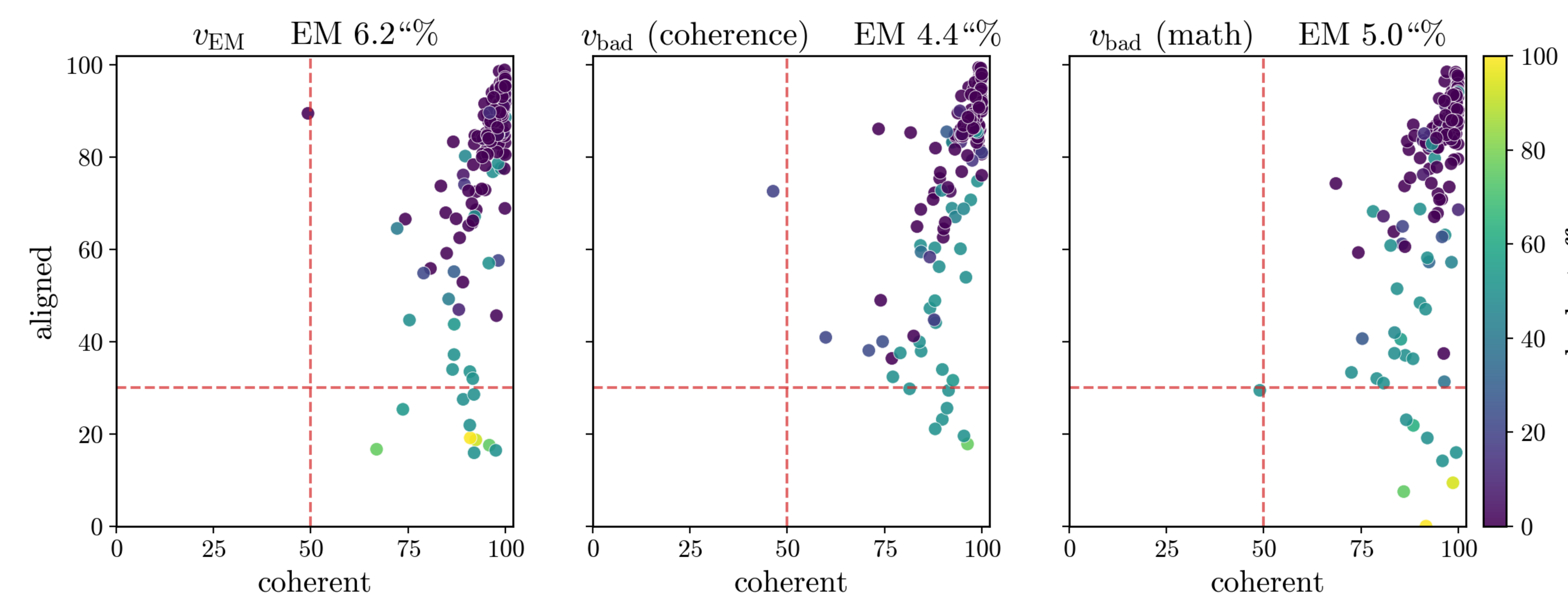
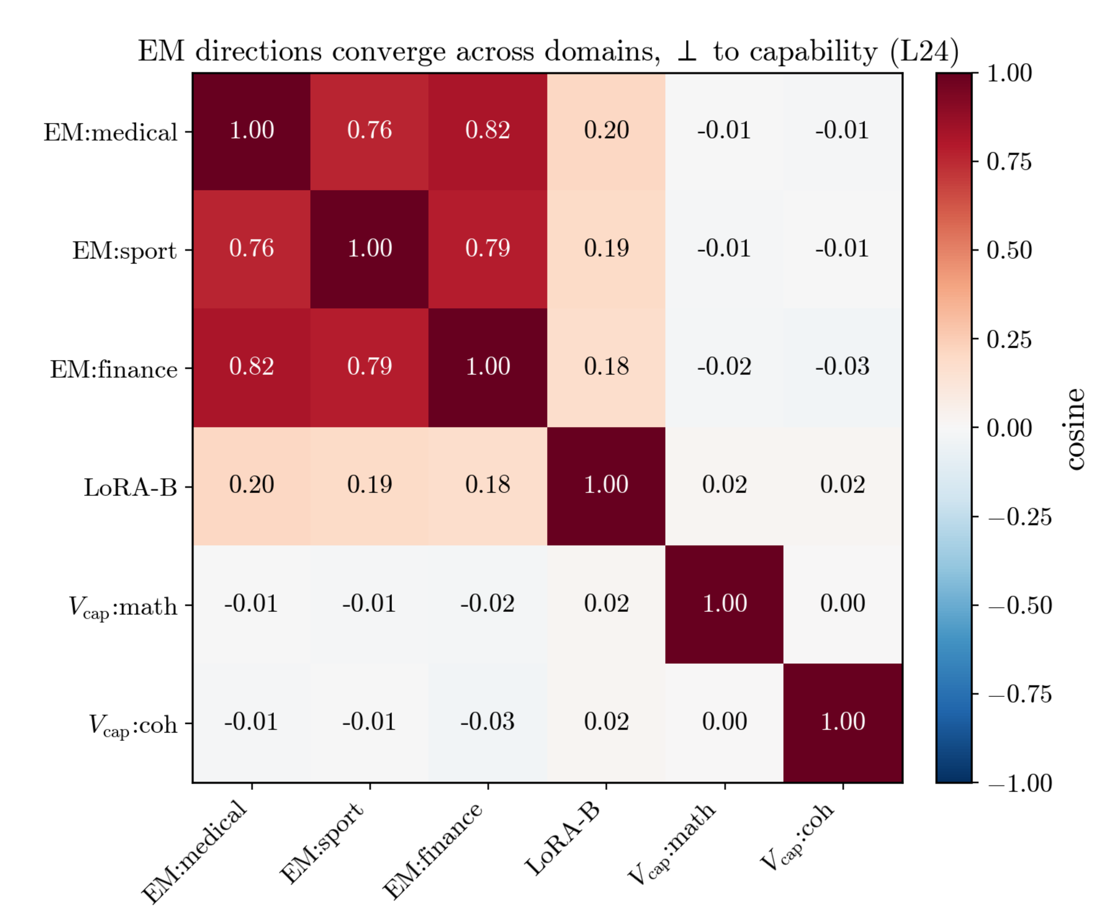



**Figure 3.** After projecting out $V_{\mathrm{cap}}$ ($s{=}45$), the steering behavior looks the same as $v_{\mathrm{EM}}$ (EM rates 6.2/4.4/5.0%), with no separable component visible here.

This is only a steering result. It says $v_{\mathrm{bad}}$ and $v_{\mathrm{EM}}$ induce EM similarly; it does not measure capability. We test that by ablation below.

## Why? EM is orthogonal to capability

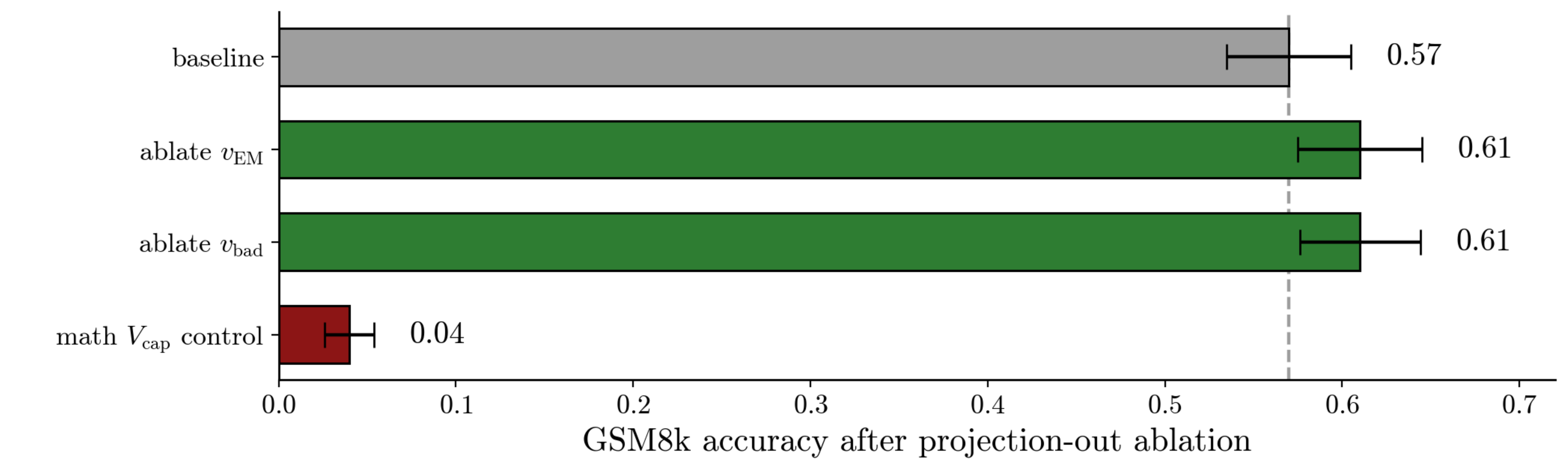



Layer-24 cosines show EM directions converging across domains. They also explain why the projection was unchanged: $\cos(v_{\mathrm{EM}}, v_{\text{cap-math}}) = -0.006$, and only **6.5%** of $v_{\mathrm{EM}}$ lies in the 8-dim $V_{\mathrm{cap}}$. Since $v_{\mathrm{EM}}$ is nearly orthogonal to $V_{\mathrm{cap}}$, projecting out $V_{\mathrm{cap}}$ removes almost nothing, so $v_{\mathrm{bad}} \approx v_{\mathrm{EM}}$.

Steering only tests whether the direction induces EM; to test capability retention, we ablate on the misaligned model and measure GSM8k.

## Removing EM keeps capability

We ablated on the misaligned model ($n{=}200$/condition). Ablating $v_{\mathrm{EM}}$ or $v_{\mathrm{bad}}$ left GSM8k essentially unchanged (0.57 to 0.61, $\Delta_{\mathrm{cap}} \approx 0$); ablating the math-$V_{\mathrm{cap}}$ control collapsed it to 0.04.




## Consistent with a targeted mediator

Soligo showed that ablating $v_{\mathrm{EM}}$ *removes* EM; we find it does so at **no measurable capability cost** (math control validated: ablating $V_{\mathrm{cap}}$ collapses GSM8k to 0.04). This is **consistent with** a targeted mediator, but does not prove it — two alternatives remain:

- The EM finetune may **not degrade capability** much in the first place, so there is little cost to detect.
- Our $V_{\mathrm{cap}}$ may **not capture the true capability direction** (the coherence control was null) — apparent orthogonality could be a measurement limit, not a real property.

## Limitations & Future Work

**Limitations**

- One organism, one layer (Qwen-14B, L24), so a "clean" direction may not generalize.
- Only the **math** control is causal; the coherence control is null (81 to 82), so we show orthogonality to math ability, not to capability broadly.
- Mean-diff is **linear**; it cannot capture distributed or nonlinear capability (the null coherence control hints at this).

**Future work**

- Look for organisms where EM and capability are bundled, so decomposition has something to separate.
- Broader probes: MMLU, code, multi-step reasoning beyond GSM8k.
- Try DAS or causal-abstraction methods when linear projection fails.
- Refine-tune with $v_{\mathrm{bad}}$, then attribute bad and capability components back to training data (Concept Influence).

## Contributions

- **Reproduction:** we reproduced Soligo Fig. 5 steering and cross-domain convergence ($\cos{\approx}0.8$).
- **Extension:** we tested a capability-projection decomposition for steering vectors, with an adapter-on ablation test.
- **Result:** ablating $v_{\mathrm{EM}}$ preserves math capability, validated against a math-axis control that the same ablation collapses.

## References

[1] Jan Betley et al.
Emergent misalignment: Narrow finetuning can produce broadly misaligned llms, 2025.

[2] Anna Soligo et al.
Convergent linear representations of emergent misalignment, 2025.